PLECTA: Overlap-aware instance reconstruction of dense strand networks from binary masks

Oday Allan[a] Igor Ostanin[a,*] Stefan Luding[a]

[a] Multi-Scale Mechanics (MSM), Faculty of Engineering Technology, University of Twente, Enschede, The Netherlands

ORCID: Oday Allan 0000-0002-6341-4695; Igor Ostanin 0000-0003-4233-4255; Stefan Luding 0000-0001-7598-0929

Abstract

Many materials and tissues are built from long, thin strands laid over one another: carbon-nanotube sheets, collagen networks, paper, nonwoven textiles. Their effective properties are governed by the architecture of the strand network, so that architecture must be measured from micrographs, with every strand recovered as a separate object that remains addressable through every crossing. Dense crossings remain challenging because intersecting strands form a single connected component, whereas signal loss can divide one strand into multiple disconnected fragments. We present PLECTA, a deterministic method that reconstructs individual strands using only a binary image in which foreground pixels mark strand centrelines. At each junction, irrespective of degree, PLECTA solves an exact maximum-weight partial-matching problem with an explicit penalty for unmatched arms. This formulation permits genuine strand terminations and produces overlapping instance layers in which crossing pixels may be assigned to multiple strands. On held-out synthetic scenes, PLECTA reconstructed strand identities more accurately than three released alternatives across all evaluated densities, and this ranking was preserved on an independently published generator evaluated using its original settings. A secondary image-informed stage uses the corresponding greyscale micrograph to estimate the depth order at each crossing and abstains when the intensity evidence does not exceed the local noise. The complete workflow is demonstrated on scanning electron micrographs of carbon-nanotube sheets, from raw image to stacked strand instances, without manually annotated images used for training or as pipeline inputs.

Keywords: strand networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

1 Introduction

Many materials are composed of long, thin strands laid over one another, some tangled at random and some deliberately twisted [1]: sheets of carbon nanotubes, collagen in connective tissue, paper, felt, nonwoven textiles, fibre-reinforced composites, aramid fibre ropes and synthetic fibre mats. The effective properties of such a material (tensile and flexural stiffness, electrical and thermal conductivity, fracture, tear and impact resistance) are governed as much by the architecture of the network as by the constituents themselves, and often more so: the stiffness and strength of a nanotube assembly are bounded by load transfer between tubes rather than by the very high modulus of any one of them [2–8]. The governing descriptors are therefore geometric: the length of each strand, its thickness, its orientation, its local curvature, and its connectivity to the strands around it. The same descriptors are the measurement target well outside materials science. A DNA fibre spreading assay reads the length ratio between two

*Corresponding author: i.ostanin@utwente.nl

labelled replication tracts along an individual fibre [9], and time-resolved imaging of the actin cytoskeleton follows how its filaments reorganise inside a living cell [10].

Although micrographs directly visualise strand networks, extracting quantitative network descriptors remains challenging. Routine image analysis measures the network in bulk (what fraction of the frame is covered, which way the texture points [11–13]) and does not resolve individual strands. To measure a distribution of strand lengths, or to ask how often one strand crosses another, every strand must first be recovered as a separate object, addressable from one end to the other.

That is an instance-segmentation problem with an unfavourable geometry: the objects are elongated, open curves whose identities must be inferred from directional continuity rather than from shape or size. The consequence is a pair of failures of opposite character. Crossings create false connectivity between distinct strands, since a crossing produces the same connected component that a genuine merge error would; signal gaps fragment one strand into disconnected components that are locally indistinguishable from genuine ends. Resolving both failure modes requires determining which pairs of arms represent continuous strands at each junction. This paper addresses that step: the reconstruction of individual strands across crossings and gaps from a binary strand-axis mask.

The essential original component of PLECTA is degree-general exact maximum-weight *partial* matching at each junction, with an explicit penalty for leaving an arm unmatched: junctions of any degree are solved by one formulation, and a strand may genuinely terminate rather than receive a forced continuation. From a binary mask alone, that matching is integrated with globally gated bridging of mask gaps, iterative re-fitting of tangent and curvature frames along the assembling chains, and overlapping instance output, in which a crossing pixel may be assigned to multiple strands. The individual ingredients, among them skeleton graphs, junction excision, continuation angles and combinatorial optimisation, are established; to our knowledge no prior method combines them, and Section 2 substantiates that claim.

PLECTA assumes that directional information remains reliable over the spatial scale of a junction: the evidence it reads is the tangent, the turns onto the chord between two tips, and the continuity of signed curvature, each estimated over tens of pixels of arclength. The assumption holds for semiflexible networks such as nanotube bundles, actin, collagen and synthetic fibres, and the method is not applicable where tangent correlation is lost within the junction scale, as for freely flexible chains; Section 5 returns to this limit.

We name the method PLECTA, after the Latin *plecta*, a braid. PLECTA is deterministic in the sense that identical input masks produce identical outputs: every parameter is fixed in a published configuration and the implementation uses no random number generator.

Application to real micrographs requires one additional component, and the workflow is divided in two. An upstream route turns a raw SEM micrograph into a binary strand-axis mask by a synthetic-data-driven CycleGAN/nnU-Net whose training pairs are synthesised rather than annotated [14]; PLECTA comprises only the reconstruction stages downstream of mask generation. This separation enables independent evaluation of instance grouping, against references whose identities are known exactly, and prevents width rendering from affecting the identity metrics. We evaluate PLECTA on 128 procedurally generated scenes spanning 20 to 60 % areal coverage, render a further 50 scenes once from a clean axis and once from a degraded one so that the cost of degradation is measured on unchanged geometry, and demonstrate both stages together on 3 annotated CNT-sheet SEM fields.

Depth ordering constitutes a separate task: determining which strand is uppermost at each crossing. PLECTA estimates this order non-destructively from the corresponding greyscale image, using the continuity of each strand's own brightness: the upper strand keeps the appearance it has on either side of the crossing, while the lower strand's is overwritten by the strand above it. The stage abstains where that difference is no larger than the local image noise. To our knowledge this is the first depth decomposition of a strand network from a single micrograph; it is reported as a demonstration with a measured accuracy, not as a validated three-dimensional reconstruction.

In evaluation, PLECTA preserved strand identity more accurately than the compared alternatives across held-out synthetic scenes at all tested densities; performance decreased with centreline density and remained sensitive to the quality of the upstream mask. The image-informed extension recovered depth order reliably where sufficient local intensity evidence was available, and abstained otherwise.

The remainder of the paper is organised as follows. Section 2 reviews the neighbouring branches of the literature and places the present method among them. Section 3 describes the synthetic and SEM data, the reconstruction itself, and the evaluation protocol. Section 4 reports accuracy on held-out synthetic scenes, a comparison with four published methods on identical masks, ablations, the depth stage, and a proof of concept on real SEM fields. Section 5 states conclusions, scope and limitations.

2 Related work

Prior work relevant to the problem of Section 1 can be grouped into five categories: network-level measurement tools, greedy instance-reconstruction methods, joint neighbourhood-optimisation methods, complete micrograph-to-measurement workflows, and monocular depth-ordering methods.

2.1 Network-level methods without instance reconstruction

Existing methods address complementary aspects of the problem. Semantic segmentation is mature, since U-Net and its self-configuring descendants reliably separate strand foreground from background [15, 16], including in materials imaging [17, 18], and nanotube micrographs have their own trained segmenters [19–21]. But a foreground mask encodes no identity, so it inherits both failures rather than resolving them. Instance methods assume the objects are compact, in materials microscopy [22] as in the general-purpose families: bounded blobs, centre-directed flows, star-convex outlines, none of which describes a long open curve that may overlap another [23–25]. Ridge and vesselness filters sharpen curvilinear signal but assign nothing across a junction [26–28], and the established fibre-analysis packages return network statistics rather than resolved objects [29–33]. Where the imaging is volumetric, strands that appear to meet in projection occupy disjoint volumes, and that separation is exploitable: in X-ray tomography of fibrous solids and of twisted ropes, individual fibres are separated by seeded watershed and morphological reasoning on the reconstructed volume [8, 34]. In a projection this separation is unavailable, because two crossing strands genuinely share the same pixels.

2.2 Greedy continuation from a mask

Recovering strands through crossings has been addressed repeatedly. Existing methods differ primarily in the evidence used for continuation and in whether earlier pairings can subsequently be revised.

A large class of methods follows a common procedure: skeletonise the mask, remove the junction pixels, and pair the resulting arms according to directional continuity. Each pairing is accepted as soon as it is scored. SIFNE reads its tip directions from the excised stubs and recovers dense microtubule networks from the mask alone [35]. Wegmayr et al. pair skeleton branches by lowest mutual angle [36]. GraFT covers a Frangi-filtered skeleton greedily, longest path first [10]. This is inexpensive, and where crossings are rare it suffices. Its weakness is structural: once an arm is paired, a later contradictory candidate cannot revise the pairing.

Two remedies have been tried. The first is to incorporate additional image evidence beyond the mask. The stronger of Wegmayr et al.'s two variants drops the angle rule and learns pixel embeddings on patches around each junction; it reads the greyscale image, and it must be trained on crops in which every fibre is already labelled individually. Even then it recovers tangled fibres at about half the precision of isolated ones, in data where two thirds of fibres cross nothing at all. Methods that pair ends with an orientation network, detect fibres with a region-based detector, or trace them recurrently [37–39] need the same labelled training data. Reading a mask and nothing else avoids that cost, a trade-off adopted in this paper. DNAi also uses only the mask but evaluates a larger set of candidate pairings, enumerating every pairing a junction allows from its endpoint angles; it does so exhaustively at degree three and four, and not at all above [9]. The closest single prior method to PLECTA is that of Liu et al., which likewise reads a mask, reconnects across gaps and permits overlapping output, but pairs termini greedily and depends on a trained orientation network [37].

2.3 Joint neighbourhood optimisation

A second remedy is to stop deciding one pairing at a time. Basu et al. and DeFiNe both weigh a whole neighbourhood at once, which is what a dense node requires. Basu et al. delete every branch point from a spanning tree over centreline points, then reassemble the pieces with an integer program that penalises each join by its bending, iterating until the assignment converges; a join that a later fit contradicts can therefore still be undone [40]. DeFiNe covers an already-built weighted graph with the smoothest paths it can find, and lets two paths share an edge, so a crossing pixel can belong to both strands instead of being awarded to one [41]. Neither, however, starts from an axis mask: Basu et al. require image intensities, whereas DeFiNe requires a graph produced by an upstream stage. Both also reassemble at most three pieces at a time, which limits how much of a fragmented neighbourhood a single round can repair. Deciding broadly and starting from a mask have so far been alternatives.

One method combines both properties, and comes closest to the present approach. De et al. reason globally and do start from a mask. Branches and junction contacts become a graph, and labels spread across it so that a decision at one node constrains its neighbours rather than resting on that node's own angles [42]. The labels, however, require initialisation. In their images they start at anatomical roots, the stained cell bodies of neurons or the optic disc of a retina, and the method returns as many strands as there are roots. A tangle of open-ended strands offers no such starting point, and how many strands it holds is precisely the unknown. Reasoning globally from a mask alone, with no seed and no strand count supplied in advance, is the problem addressed in this paper. Four of these methods are measured against PLECTA directly in Section 4.5, on identical masks and through one scorer. Skeleton graphs, junction excision, continuation angles, distance gates, combinatorial optimisation, iterative refitting and overlapping output are therefore all established; none is claimed as new in this paper.

2.4 Complete workflows, and depth from a single view

The closest complete workflow is that of Peters et al., who join a U-Net-like segmentation trained on over a thousand annotated SEM micrographs to algorithms that trace fibres through intersections, and validate against human evaluators on fibre number, length and width [43]. Both workflows separate semantic segmentation from instance reconstruction but evaluate the reconstruction stage differently: Peters et al. assess the replacement of human counting and measurement, whereas PLECTA is isolated behind a binary mask and evaluated on the preservation of strand identity across crossings, against overlap-aware references in which every pairing is known.

Which strand lies on top at a crossing is a further question, and for nanostructured networks it is normally answered destructively, by focused-ion-beam sectioning and tomographic reconstruction [44], because a micrograph's intensities record surface topography rather than height. Monocular depth ordering has long derived pairwise depth from local occlusion evidence, assembled those relations into a depth-order graph and corrected the contradictions among them [45, 46], and single-view hair modelling resolves the layering of intersecting strands by analogous reasoning [47]. Those methods read the shape of the outlines where two objects overlap, which a bundle a few pixels across in a noisy micrograph does not reliably provide; the evidence used in this paper is instead the continuity of each strand's own brightness across the crossing.

3 Materials and methods

This section defines the method's inputs and outputs, describes the synthetic and experimental data, presents the reconstruction and the image-informed stages, and specifies the evaluation metrics.

3.1 Scope and workflow

The input is a binary strand-axis mask $M \in \{0,1\}^{H\times W}$, whose foreground $F = \{x : M(x) = 1\}$ is one pixel wide and traces strand centrelines rather than strand widths. From it PLECTA reconstructs an overlap-aware collection $\{I_1, \ldots, I_N\}$ with $I_n \subseteq F$, so that every emitted pixel is one the mask already contains, and a crossing pixel may belong to several strands at once. The evaluated grouping stage uses only M; where a greyscale SEM image exists it is read downstream, for width and brightness, and cannot revise the fixed grouping.

Figure 1 defines the scope of PLECTA within the complete workflow. The method uses only mask geometry; the junction and gap matchings are recomputed over eight rounds, with frame parameters refitted from the currently assembled chains before each round. Mask generation is described in Section 4.3.

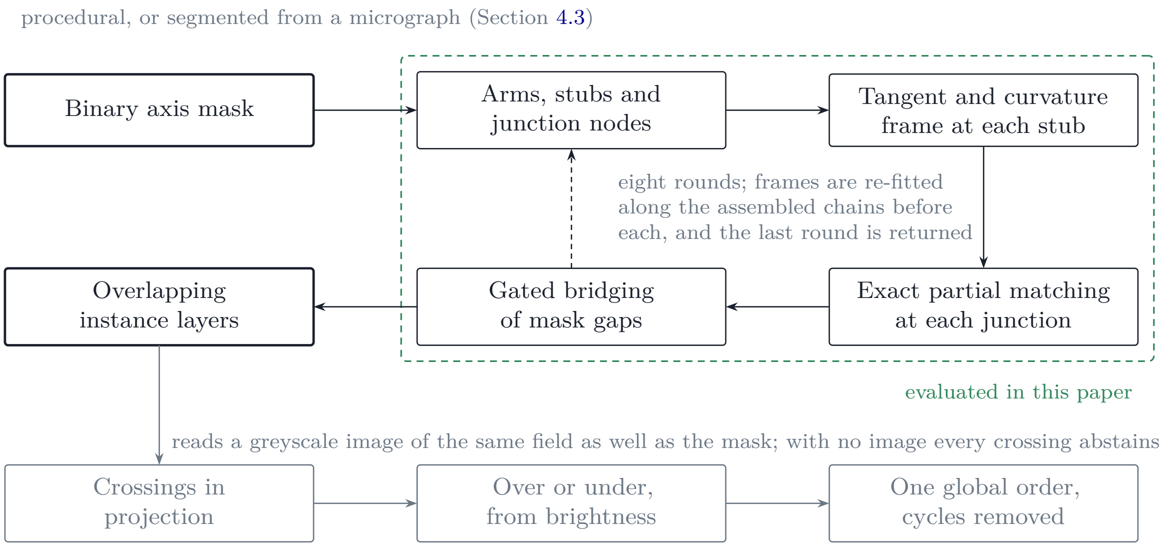

Figure 1. Scope of PLECTA within the complete workflow. A single binary axis mask is the method's only input, and every result in this paper measures the stage inside the dashed boundary, which maps one binary mask to a set of overlapping instance layers. The two matchings are re-made over eight rounds as frame estimates improve along the assembled chains, and the last round is returned. The depth stage reads a greyscale image of the same field in addition to the mask, adds a third dimension without revising any identity, and abstains at every crossing when no image is supplied.

The central difficulty is the single crossing: the mask contains one connected component from which several arms leave, and no local evidence states which arm continues into which. The output is a set of layers rather than a partition, so a pixel inside that component is counted in both strands that contain it.

3.2 Synthetic data and evaluation split

Because the scenes are generated rather than annotated, the reference identities are known exactly. For every strand k the generator records its ordered centreline $\mathcal{C}_k$ and the pixels it was drawn onto, its support Ω_k. The supports are kept one per strand instead of being merged, so a crossing pixel belongs to both strands that pass through it and the reference is overlap-aware by construction.

Each scene is written out four ways: the exact one-pixel axis; that axis degraded with small gaps and raster irregularities; the strands at full width, merged into a single foreground; and an SEM-like greyscale image, in which each strand carries its own brightness and a slow variation along its length, blurred by one point-spread function over a low-frequency background with grain. A depth-resolved variant of the generator supplies the missing structure for the depth stage of Section 3.5.2. The generator assigns each strand a height and a radius, enforces non-interpenetration, and renders the stack from the top down, so the image exhibits occlusion, defocus that grows with distance from a randomised focal plane, and width broadening. Those cues are randomised independently of one another, so no single rule such as brighter-is-higher recovers the order.

Scenes are labelled throughout by areal coverage, the fraction of the frame the full-width strands cover rather than the fraction the thin axis occupies. Coverage serves as a proxy for scene difficulty but does not directly affect either the input or the metric, both of which operate on centreline geometry. Under this generator, coverage correlates with centreline length density, and Section 4.1 separates the two effects.

Every parameter was fixed on a separate development set and never on the scenes reported here; the 128 evaluation scenes, 32 at each of 20, 30, 40 and 60 % coverage, use seeds that appear nowhere in that development set. Supplementary Section S6 gives the seed blocks and the verifier that checks their disjointness.

A separate controlled paired set of 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage, was scored once from each geometry's clean axis and once from its degraded mask, after the configuration was fixed.

3.3 CNT SEM data and mask sources

Three manually annotated CNT-sheet SEM fields were available, each 1536 × 1024 px: B58_100 (307 visible CNT bundles), B58_110 (430) and B58_300 (325). The annotated strand is the visible bundle, not the individual nanotube, which SEM does not resolve at this magnification. The nominal horizontal field width is 1.66 µm across the long, 1536-px axis of the stored images, which is 1.08 $\mathrm{nm\,px^{-1}}$. Each annotation is one human interpretation of projected, locally ambiguous crossings.

The reference annotations were drawn with an in-house interactive instance-segmentation tool for greyscale SEM of thin, crossing strands. Its data model is per-instance masks rather than a flat label image, so a pixel may belong to several strands at once, which makes the reference overlap-aware and comparable with PLECTA's output. The file the tool saves is the file the evaluation reads, with no export or conversion step in between. The annotator paints each visible bundle as its own strand, with two forms of assistance: strokes may snap onto the local intensity ridge, with polarity calibrated from the existing annotation, and may be re-rendered through a smoothing-spline fit whose centreline is shifted onto the measured intensity crest only where that crest is consistent. Widths are measured from perpendicular full-width-at-half-maximum profiles, referenced to the local minima on either side of the peak and reported as twice the nearer half-width, so a crossing strand fusing into the profile does not bias the measurement wide. Candidate gap bridges, collinear endpoint pairs of different strands, are suggested but never applied automatically; each is accepted or rejected by the annotator, so identity through every crossing is a human decision.

The following limitations apply to the annotation protocol. The reference annotations were drawn by the first author (O.A.), hereafter annotator A. These are manual references, not the ground truth. A second annotator, B, who is not an author, has annotated 3 of these fields independently, and Section 5 reports what that measurement shows: the two annotators do not implement one strand definition, so every real-data score in this paper is reported against a named annotator and the two are never averaged. Scores throughout are computed on centreline fragments, not on strand widths.

Two inputs per field were derived. The first is the skeleton of annotator A's strand annotation, an oracle-like condition that asks what the grouping can do when the axis evidence is curated. The second is an automatic mask from the upstream CycleGAN/nnU-Net segmentation of Section 4.3 [14, 16], which asks what an automatic mask allows it to do. That segmentation is a separate contribution with its own evaluation; all that matters here is that its output is a binary mask produced without seeing any annotation.

Mask quality was compared after skeletonisation with the symmetric 2-px tolerance used throughout, which avoids rewarding mask thickness, and reconstruction scores were computed within each mask's own visible common-fragment population.

The fields were acquired on a Thermo Fisher Helios 5 UX DualBeam with an Elstar field-emission column in immersion mode, imaged with the through-lens detector at 2.00 kV and 400 pA and a working distance of 4.3 mm. Each frame is an integration of 35 passes at 100 ns dwell with drift correction enabled.

3.4 PLECTA reconstruction

This section describes the method itself. It reads the binary axis mask and nothing else, and the subsections follow the order of execution: the skeleton graph derived from the mask, the pairing cost, the junction matching, the gap bridging, and the iteration schedule.

3.4.1 Skeleton graph and stub frames

PLECTA skeletonises the mask using the Zhang–Suen thinning algorithm [48] and removes the short spurs skeletonisation produces. Skeleton pixels where three or more paths meet form junction clusters, and clusters close enough together are merged into a single *node*: at this resolution a crossing is an extended cluster rather than a point, and neighbouring crossings often touch. What remains between nodes is an *arm*, an unambiguous run of strand, carrying a *stub* at each end.

Each stub is given a frame comprising its tip position, outward tangent and curvature, estimated over a short arclength interval behind the tip. A stub whose support is too short to yield a reliable direction is marked as unreliable rather than estimated. Once arms are linked into chains, the frames are re-estimated over the expanded chain-level support, which is why the method iterates rather than deciding once.

3.4.2 The pairing cost

The pairwise linking cost comprises three geometric terms (Figure 2): the reversal between the two outward directions, which vanishes when one strand continues into the other; the turns from each direction onto the tip-to-tip chord, which penalise parallel arms that run past one another; and the continuity of signed curvature, which penalises joins that would require a kink. For gap candidates the tip separation is charged as well; the distance term is omitted for junction links. The cost of linking stubs a and b is therefore

$$C_{ab} = w_d\,\theta + w_t\,q(d)\,(\varphi_a + \varphi_b) + w_\ell\,\frac{d}{\ell_0} + 30\,w_\kappa\,|\kappa_a + \kappa_b|, \qquad q(d) = \min\left(1, \tfrac{d}{d_0}\right), \tag{1}$$

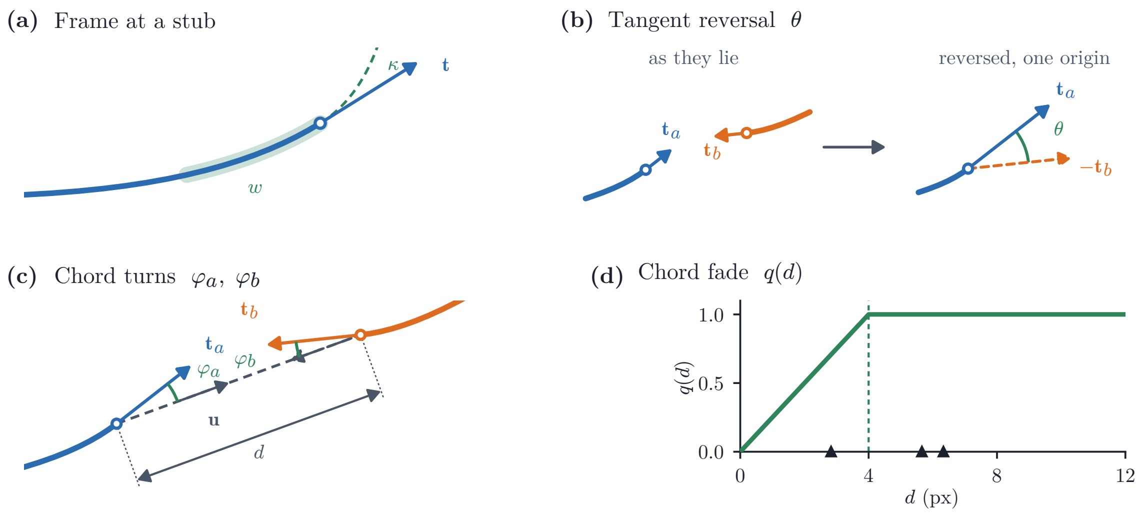

Figure 2. The stub frame and the terms of the pairing cost. **(a)** The least-squares arclength fit over window w, read outward from the tip at $s = 0$, giving outward tangent $\mathbf{t}$ and signed curvature κ; quadratic when the span allows, linear otherwise. **(b)** θ, in the two steps its definition takes: the outward tangents where they lie, then the same pair with $\mathbf{t}_b$ reversed and both read from one origin. A perfect continuation makes them antiparallel, so $\theta = 0$. **(c)** The chord turns φ_a, φ_b onto the chord $\mathbf{u}$ joining the tips. **(d)** The fade $q(d)$ against tip separation; triangles mark the six real separations at the crossing of Figure 3.

with d the tip separation, θ the reversal angle between the two outward tangents, and φ_a, φ_b the turns from each tangent onto the tip-to-tip chord, measured at their own vertices; junction links set $w_\ell = 0$.

Three properties of that expression are deliberate. The reversal angle ignores the chord, which keeps it meaningful when the tips are three pixels apart and the chord direction is quantisation noise. The chord turns do see lateral offset, so two parallel arms that never meet are penalised. And because it is the chord terms that degrade over short separations, the fade $q(d)$ withdraws them as the tips approach rather than letting an unstable angle dominate. The weights, the scales ℓ_0 and d_0, and the constant placing curvature on the scale of the angular terms are fixed parameters, listed in Supplementary Section S1.

3.4.3 Candidate gates and exact matching

Leaving a stub unmatched carries a penalty rather than a prohibition, and that penalty serves two purposes: it also fixes what may be included as a candidate at all, since an edge costing more than the two penalties it would save could never be chosen. Exact maximum-weight partial matching then picks, over disjoint pairs,

$$\max \sum_{(i,j)\in\mathcal{M}} (p_i + p_j - C_{ij}), \qquad \text{admissible iff } C_{ij} < p_i + p_j, \tag{2}$$

which is the same as minimising the total link cost plus a penalty p_i for every stub left unmatched. Junction and gap links carry different penalties, and a gap link must additionally clear limits on how far it may reach and how far either direction may turn, the tighter set, because a gap asserts a continuation across evidence that is *missing* rather than merely ambiguous. Supplementary Section S1 lists the penalties, the limits, and the schedule below.

The two matching stages are applied iteratively, over several rounds, with every frame re-fitted along the chains assembled so far before each round. Early rounds are deliberately conservative: penalties are scaled down, which admits fewer edges, and the gap angle limits are narrowed, so only pairings that survive the stricter test are committed while the frames are least reliable. The constraints relax towards their configured values as the assembling chains improve the frame estimates, and only the final round runs at the configured parameters. That round is the one returned.

Every stub is therefore either paired or left unmatched at the stated penalty, and there is no third possibility. That option is what permits a genuine strand termination, and what prevents a node with an odd number of arms from forcing a continuation onto the one left over. No node is classified: a node of any degree is solved by the same matching, and an arm ends because the unmatched state maximised the objective, not because the node was recognised as a particular kind.

It also removes a parameter rather than adding one. Admissibility in Equation 2 is not a threshold chosen alongside the penalty: an edge costing $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching could not select it under any setting, and excluding it from the candidate set removes only edges that could never have been chosen. One number therefore states both the penalty for an unmatched arm and what may be considered, where a method that gates separately must set a cost threshold and a penalty and keep them consistent.

Solving a junction as one matching rather than one edge at a time is the distinction claimed in Section 1, and Figure 3 shows what it achieves on real nodes, alongside the other two decisions the linker makes. A locally attractive edge can be declined, as at the three-arm node of Figure 3b. Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense: they never disagree at nodes of degree two to four, disagree at roughly a fifth of nodes with five to eight stubs, and at two thirds of nodes with nine or more, where the joint solution is the cheaper one; Supplementary Section S1 gives the counts. That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, an early low-cost selection propagates and the difference is no longer confined to dense nodes.

Two measurements evaluate that claim. The greedy continuation baseline of Section 4.4 reads the *same* masks and the same local geometry and differs only in pairing greedily instead of jointly, so the +0.145 F_1 between them isolates the formulation rather than the evidence. The component ablations reported alongside it then ask the complementary question, whether the rest of the combination contributes measurably, and find that removing the junction chord-turn term costs -0.1858 and removing gap matching -0.1394. The two measurements are complementary: the first evaluates the joint formulation itself, and the second whether the terms it matches on are redundant.

3.4.4 Strand output

Connected components of accepted arm links define strands. PLECTA paints every arm pixel and every junction cluster touched by a chain into that strand, so intersecting strands share crossing pixels. Accepted gap links determine identity but are not painted into the fixed centreline output. All parameters are fixed in a published configuration (Supplementary Section S1).

3.5 Image-informed reconstruction

The grouping above reads the mask alone and is what this paper evaluates. Where a greyscale image of the same field exists, three further stages can read it: two give each strand a width, and one orders the strands at their crossings. All three run after the grouping is fixed and none can move a pixel from one strand to another, so no identity score in this paper depends on any of them. All are governed by fixed parameters (Supplementary Section S1), and none is specific to SEM: they take whatever image accompanies the mask, and SEM is simply the case demonstrated here.

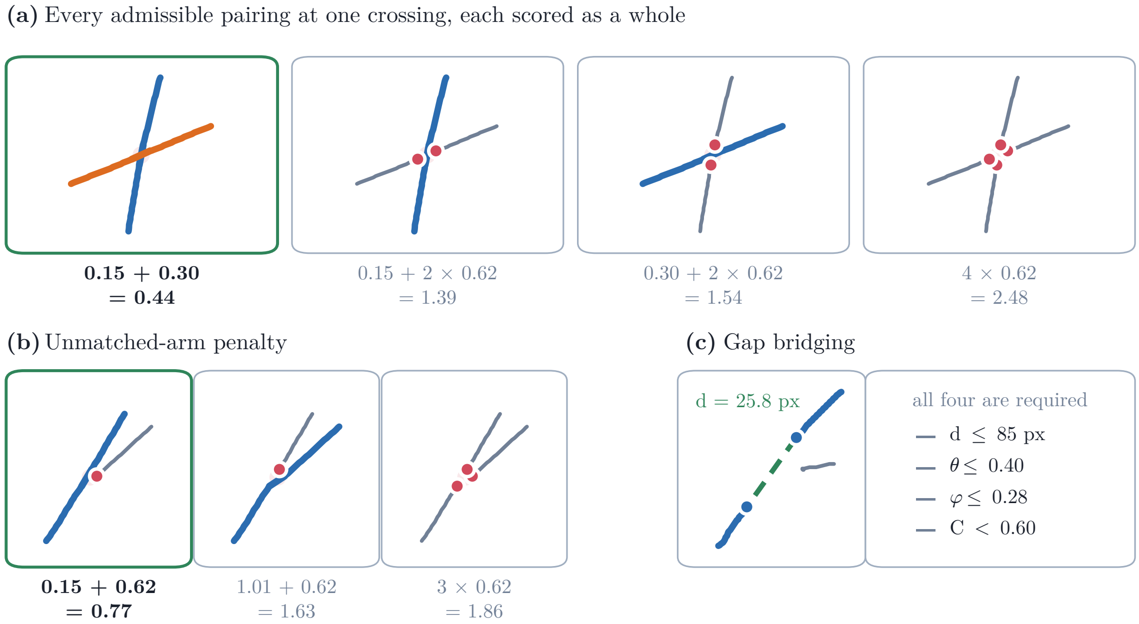

Figure 3. Candidate-link evaluation and matching outcomes, on the nodes where the decisions were made. Every arm is the skeleton the numbers were measured from, so every angle shown is the one the cost was computed on. **(a)** Every pairing the admissibility gate allows at a four-arm crossing, each scored as a whole: its edge costs plus the penalty $p_i = 0.62$ of each arm it leaves unmatched. The green frame marks the one returned; two further pairs exceed $p_i + p_j = 1.24$ and are excluded as candidates. **(b)** A three-arm node where an admissible edge is declined on the total rather than on its own cost. **(c)** A gap link, which must clear all four gates at once.

3.5.1 Width and ribbon rendering

Perpendicular intensity profiles, sampled along each ordered chain away from junctions, attach an apparent width, a brightness, an uncertainty and a quality flag to an existing strand without altering its arm membership. Ribbon rendering then interpolates accepted gaps, resamples and smooths the centreline, and draws the chain at its fitted width, excluding junction blobs. Neither silhouette absorption, which would union rendered strands, nor any other step merges two strands after the fact, and none contributed to the held-out result.

3.5.2 Depth decomposition

A projected crossing belongs to both intersecting strands but does not encode their relative depth. The depth stage estimates that order.

Evidence at one crossing (Figure 4). Where two strand centrelines intersect, a short arc of each is taken through the crossing (the shared *core*, a fixed length, the same at every crossing), together with the *flanks* on either side of it, kept clear of the other strand's stroke so that they measure one strand and not the union. One channel is read on both: median intensity. The physical argument is occlusion. Whichever strand lies on top presents the same surface through the crossing as it does beside it, so its core matches its own flanks in brightness; the lower strand is overwritten there, so its own continuity breaks. For each strand, the difference between the pooled core intensity and its own flank intensity is computed, and the strand with the smaller difference is assigned as upper. The score s quantifies the difference between the two matches, normalised by the larger of the inter-strand brightness separation and the noise measured at that crossing. No order is assigned when $|s|$ falls below a fixed threshold: the crossing *abstains* rather than being decided by noise. Dividing by the noise, rather than by the separation alone, is what makes a confident score mean *separated by more than the noise* rather than merely separated.

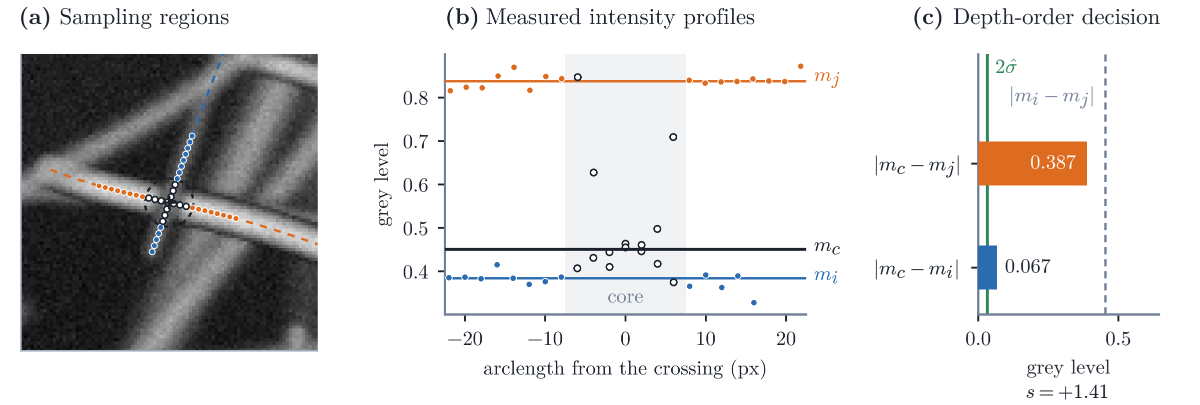

Figure 4. What the depth stage measures, at one crossing where two strands meet at 83.8°, chosen as a legible example rather than at random. All positions and grey levels were obtained directly from the stage output. **(a)** The core arc shared by the two strands, and each strand's own flanks, which the window keeps clear of the other's stroke; open dots are core samples, filled ones flanks. **(b)** The same samples against arclength: the pooled core median m_c sits by strand i's flank level and far from j's, which is the occlusion. **(c)** The two distances the score compares, over the larger of the strands' contrast and twice the flank noise. That ratio is the score s of Supplementary Section S2, whose sign names the upper strand and whose size weights the crossing below.

One order for the whole field. Local decisions need not be mutually consistent: they can assert that A is above B, B above C and C above A, which no assignment of heights satisfies. Each decided crossing therefore contributes a directed edge weighted by $|s|$, the strength of its own evidence (Figure 5). Within each connected component of the crossing graph, one total order over its strands is estimated by maximising the total weight of satisfied edges, which is equivalent to reversing the minimum-weight set of relations that leaves no contradiction: the minimum-weight feedback arc set objective. The objective is solved exactly for components small enough to search and greedily for larger ones. All pairwise relations are then derived from the component-level order, so even a relation not involved in any local contradiction may be reversed. Each reversal and its original local score are retained, to quantify how far the global reconciliation moved. Depth propagates only along crossings, so two strands in different components are related by no evidence at all and their order is a stated tie-break, not a measurement. Supplementary Section S2 gives the sampling geometry, the score in full, the measurement that replaced an earlier two-channel rule with it, and the solver.

Layers and height. The corrected relations form a directed acyclic graph whose longest path fixes the smallest number of layers consistent with them; strands that never conflict share a layer, so the layer count follows from the ordering rather than from the number of strands. Each strand is then placed at a height that respects its order and keeps strands from interpenetrating at their measured radii, with the stack compacted so that the film is as thin as those constraints allow.

Without a greyscale image no local depth relations are estimated and the predefined tie-break determines the output order: the stage requires image evidence and reports how much of it was usable. Section 4.2 evaluates it, and Supplementary Section S5 defines the measures used, which are disjoint from those above because none of them can see depth.

3.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1. The input mask is cut into fragments, each one unambiguous arm between crossings, by a rule that reads neither the reference nor any

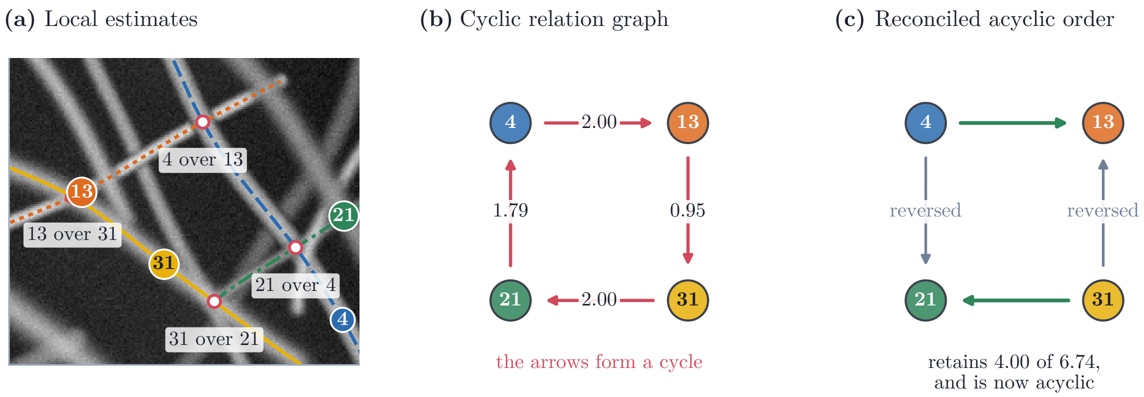

Figure 5. Global reconciliation of local depth-order estimates, on a contradiction encountered by the stage. **(a)** Four strands and the four local estimates made at their crossings. **(b)** Those estimates as a graph: each arrow points from the strand called upper to the one called lower, and following them returns to where it started, so no stack of heights satisfies all four. **(c)** The order satisfying the most evidence by weight retains the two heaviest estimates and reverses the two lightest, which are recorded rather than discarded. Layer numbers are not shown: a strand's layer follows from its whole component, and these four sit in a scene of seven.

prediction, so every method compared is cut into the same pieces; each fragment then takes a reference and a predicted identity by one rule applied to both sides alike, and the score counts the pairs of fragments the prediction groups as the reference does. Figure 6 illustrates the endpoint on the smallest case that contains a crossing, including the two conditions under which it is undefined rather than zero.

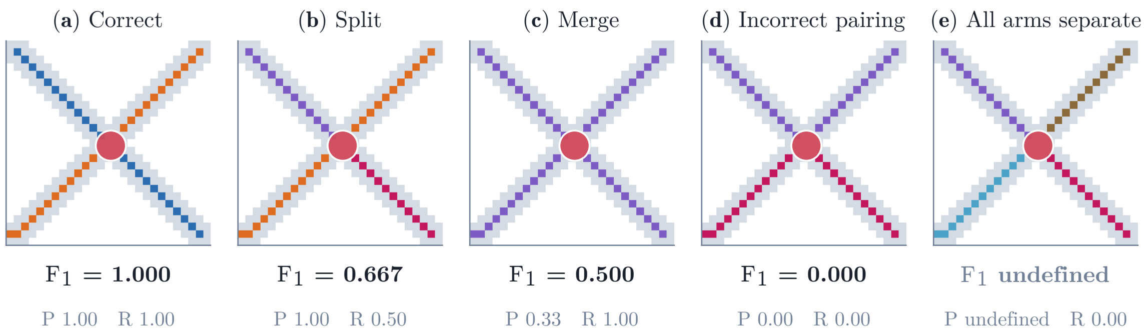

Figure 6. Illustration of the endpoint metric on the smallest case that contains a crossing. The scorer removes a neighbourhood of the junction (red) and evaluates, for each of the six pairs of the four remaining arms, whether the grouping joins it as the reference (a) does. Splits (b) reduce recall and merges (c) reduce precision; the all-separate grouping (e) produces no pair, so precision and F_1 are undefined rather than zero, one reason the adjusted Rand index and variation of information are reported alongside.

Supplementary Section S3 gives the construction and the assignment rule in full. Because junction pixels are deleted before fragments are labelled, the measure is insensitive to whether an emitted crossing pixel is given one owner or two; it registers only whether the arms meeting there are grouped as the reference groups them.

No single metric is quoted alone, because the metrics are sensitive to different error types: pairwise F_1 evaluates relationships between fragments and is quadratic in fragmentation, while object-detection metrics evaluate object existence and are insensitive to an arm exchange at a crossing, which they rate 1.000 where the pairwise score is 0.000. The adjusted Rand index and variation of information [49, 50] are therefore quoted beside the pairwise score, with detection F_1 and Crossing Fidelity. The input masks themselves are characterised by a tolerant recall,

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information, for which lower is better; higher is better for the other metrics.

Group	n	F_1	Precision	Recall	ARI	VI split	VI merge	VI total
All	128	0.85	0.87	0.84	0.85	0.26	0.22	0.47
20%	32	0.91	0.93	0.90	0.91	0.14	0.09	0.23
30%	32	0.87	0.89	0.86	0.87	0.21	0.16	0.37
40%	32	0.86	0.86	0.85	0.85	0.24	0.22	0.47
60%	32	0.77	0.79	0.76	0.77	0.44	0.39	0.82

precision and Jaccard index, called completeness, correctness and quality in the thin-structure literature, matched within a tolerance rather than pixel for pixel, because two skeletonisations of one strand seldom fall on the same pixels. Supplementary Section S3 defines each in full, with its equation, what it rewards and what it is insensitive to.

Supplementary Section S3 states the interval construction, the resampling unit and the cases where no inference is drawn.

4 Results

Results are presented for synthetic reconstruction, depth decomposition, real SEM images, component ablations, and comparisons with published methods.

4.1 Reconstruction on synthetic images

PLECTA is scored first where the reference identities are known exactly: the held-out synthetic scenes, under the measures of Section 3.6 and against the overlap-aware reference the generator recorded when it drew them. Table 1 reports every measure overall and by density stratum; the text then analyses the overall score, its dependence on density, and the interpretation of the density label.

Mean common-fragment pairwise F_1 over the 128 held-out scenes was 0.85 (density-stratified 95% bootstrap CI [0.84, 0.86]); Table 1 gives precision, recall, the adjusted Rand index and both variation-of-information components, overall and by density. These scenes used unseen seeds from the same procedural generator as development data, so this is same-generator generalisation and not distribution transfer (Supplementary Section S6).

Performance decreased with scene density: mean F_1 fell from 0.91 [0.89, 0.93] at 20% coverage to 0.77 [0.76, 0.79] at 60%, monotonically through the intermediate strata. The same decline appears on the paired clean-against-degraded set below, where the clean and the degraded curves separate at low coverage and meet at 50–60 %: the effect of degradation was smallest at high density, where junction ambiguity dominates gap-related error. Per-scene runtime had median 1.12 s, ranging to 34.06 s on the densest scenes; Section 4.5 measures cost systematically, where the transferable quantity is the scaling exponent rather than any absolute time.

Areal coverage labels these strata but does not cause the difficulty they report, and it cannot: PLECTA is handed a centreline mask and never a width. Rendering one geometry at three bundle widths was tested directly on 36 development scenes, and it leaves the input mask and the overlap-aware reference byte-identical while areal coverage varies by up to a factor of 2.49; F_1 moved across those renderings by 0.0000. What does move the score is centreline length density, mean F_1 falling from 0.92 to 0.84. A coverage stratum should therefore be read as a proxy for the length density accompanying it under this generator.

How much of that score the input mask fixes, rather than the grouping, is separable. On the separate 50-scene controlled set, mean F_1 was 0.82 from degraded axes and 0.84 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113,

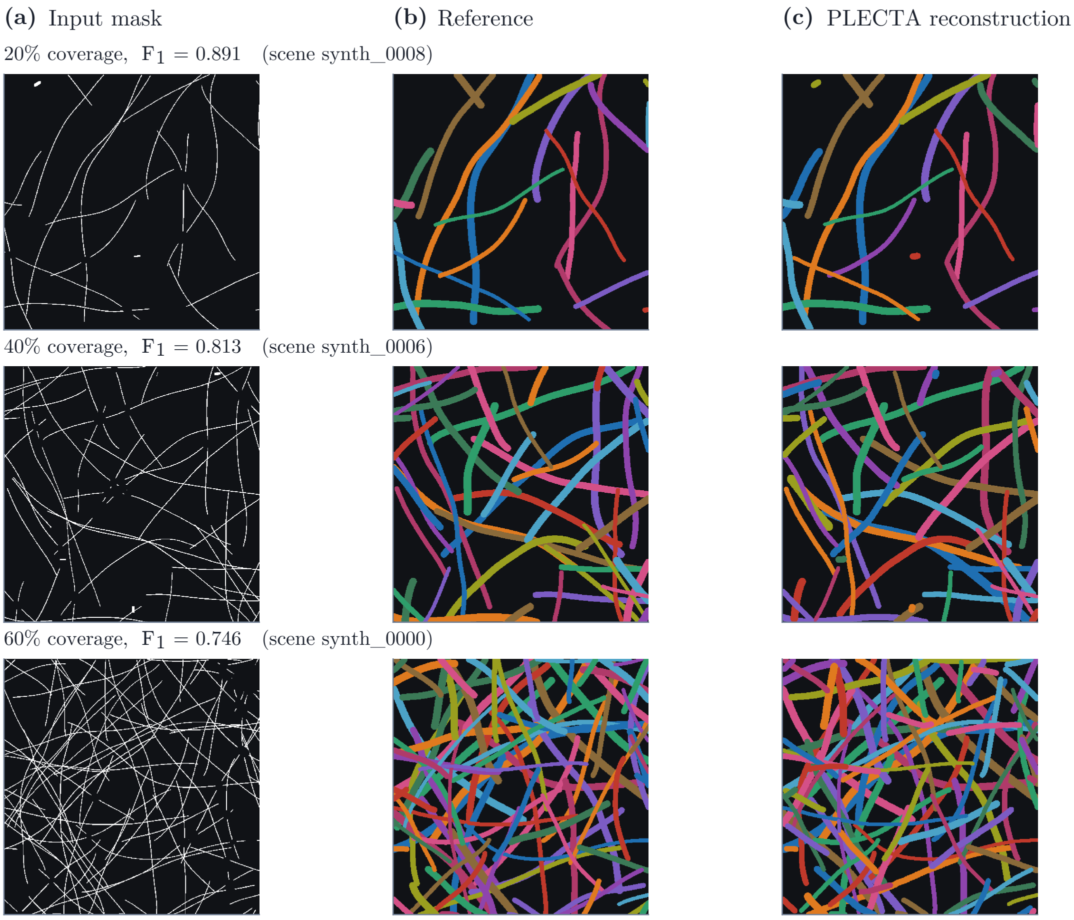

Figure 7. Reconstruction at three coverage levels, each the scene closest to the median F_1 of its stratum, shown as a horizontal window of the scene. Columns: the degraded input mask, the overlap-aware reference, and PLECTA's strands at their fitted widths. The score beside each row is the grouping's, not the rendering's. Strand colours are arbitrary and not matched between columns.

0.0333]). Modelled gaps and raster irregularities therefore caused a small resolved loss under this generator.

Figure 7 shows what the score corresponds to scene by scene, showing at each of three coverage levels the scene whose F_1 is closest to the stratum median.

4.2 Depth decomposition

The stages above reconstruct strands in projection. A crossing pixel is assigned to both strands that own it, but nothing so far says which of the two lies on top. The depth stage answers that, reading the greyscale after the grouping is fixed and leaving every strand exactly as the stages above left it. Its evidence is the continuity of each strand's own brightness across the crossing, weighed against the noise measured there; from those local estimates it imposes one globally consistent vertical order and places each strand at a height at which nothing interpenetrates.

It is evaluated on 12 scenes from the depth-resolved generator of Section 3.2, at two areal coverages, in two input conditions. Given the reference 2-D geometry, the depth decision is measured alone; given only the binary axis mask, the same stage runs on PLECTA's own reconstruction and carries its error. Table 2 reports both, under the measures Supplementary Section S5 defines. Its two blocks are not interchangeable. The crossing rows score the over/under call at each crossing, one score per crossing, taken before anything is reconciled; the stacking rows score the height field the solver returns once every call has been pooled, one score per pair of strands. Two strands that cross three times are three crossings and one pair.

Table 2. The depth stage on 12 synthetic scenes whose true vertical order is known. Chance is 0.5 for every accuracy shown.

	Given 2-D geometry		Reconstructed 2-D	
Areal coverage	20 %	40 %	20 %	40 %
At each crossing				
Projected crossings per scene	28	150	28	150
Crossings recovered	1.00	1.00	0.86	0.75
Order accuracy, all crossings	0.81	0.74	0.73	0.56
restricted to those it decided	0.89	0.88	0.96	0.91
Declined, of the crossings it found	0.08	0.16	0.12	0.18
In the stacking it outputs				
Depth order, all strand pairs	0.67	0.71	0.70	0.72
restricted to pairs that cross	0.86	0.86	0.91	0.85
Pairs within one crossing component	0.39	0.84	0.36	0.75
Layers in the reference	5.2	8.7	5.2	8.7
Layers recovered	5.2	8.3	5.3	8.0
Layer agreement	0.66	0.42	0.70	0.38

Accuracy is high where image evidence exists and substantially lower where it does not, and the table separates the two. Among crossings for which an order was assigned, accuracy ranged from 0.88 to 0.96, whether the 2-D geometry was given or reconstructed and at both crossing densities. The remaining rows report the recovery and abstention rates.

Over every pair of strands in the scene, order accuracy falls to 0.70, and that is an absence of evidence rather than a failure of the decision. Depth propagates only along crossings, so two strands that never meet, and are joined by no chain of strands that do, are related by nothing the image contains; their order is a stated tie-break, since the available crossing evidence does not determine their relative order. Only 0.36 to 0.84 of strand pairs lie within one connected component of the crossing graph, which is the fraction for which any evidence exists at all. This fraction largely accounts for the reduction. A model that orders connected pairs at the decided-crossing accuracy and assigns unconnected pairs at chance predicts 0.65 at 20 % coverage, against 0.70 measured. At the sparser coverage the stage exceeds that prediction, so the unconnected pairs are evidently not entirely uninformative; at the denser end it falls short, which is what error accumulating along the long transitive chains of a large component would do. Depth claims are therefore scoped to the connected component, and a globally ordered film would need evidence that does not travel through crossings, such as defocus, or a second view, rather than a smoothness assumption, which these scenes would not support: their heights are drawn independently, so proximity carries no depth information by construction.

All-crossings accuracy is additionally reduced by crossings the 2-D stage never recovered and by abstentions, and the table gives each its own rate. Decided-crossing accuracy must therefore be read together with the recovery and abstention rates, because increased abstention can raise conditional accuracy without improving overall performance.

Layer agreement degrades the way the global order does, falling from 0.70 to 0.38 as coverage doubles while the recovered layer count stays close to the reference: local ordering remained sufficiently accurate to recover a similar number of layers, while small order errors move strands across layer boundaries.

Figure 8 shows what the stage produces on one synthetic scene, from the binary axis mask through to the film the reconstruction implies; Figure 9 is the same sequence on a real CNT SEM field, which no depth ground truth can score.

(a) Reference image **(b)** Re-rendered **(c)** Depth reconstruction

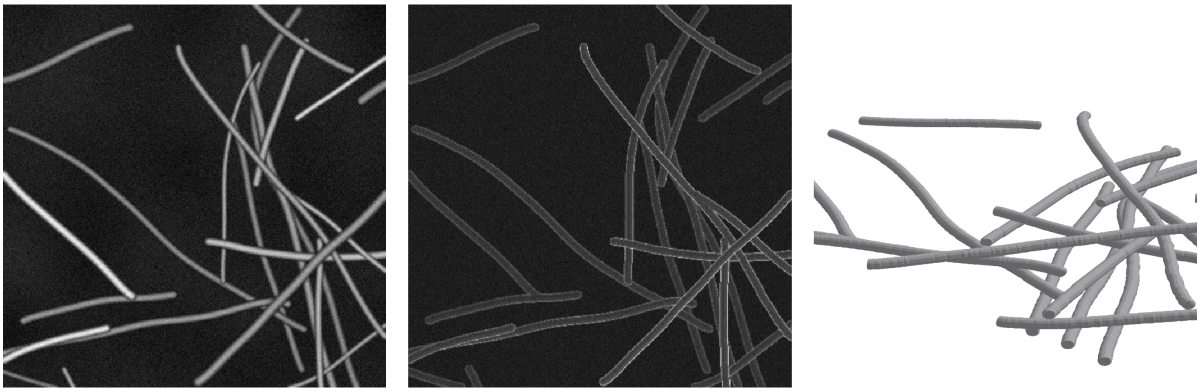

Figure 8. The depth stage on one synthetic scene, reconstructed from the binary axis mask. The image the scene was rendered from, that scene re-rendered from the reconstruction through a forward SEM model, and the film the reconstruction implies. The re-rendered panel is a consistency check on depths and radii, not a scored quantity: the difference Table 2 reports between input conditions is a difference in which crossings were *found*, which a rendered image cannot show. The scene is 512 px across, in the generator's pixel units.

4.3 Proof of concept on real SEM

Application to real SEM images requires an upstream mask-generation method; here the binary axes are supplied by the segmenter of Figure 10.

The upstream segmenter (Figure 10) ends in a binary strand-axis mask [14], and PLECTA reads that mask and nothing else. Three manually annotated CNT-sheet fields separate the two: PLECTA is run once on the skeleton of annotator A's annotation and once on the nnU-Net

(a) Micrograph **(b)** Re-rendered **(c)** Depth reconstruction

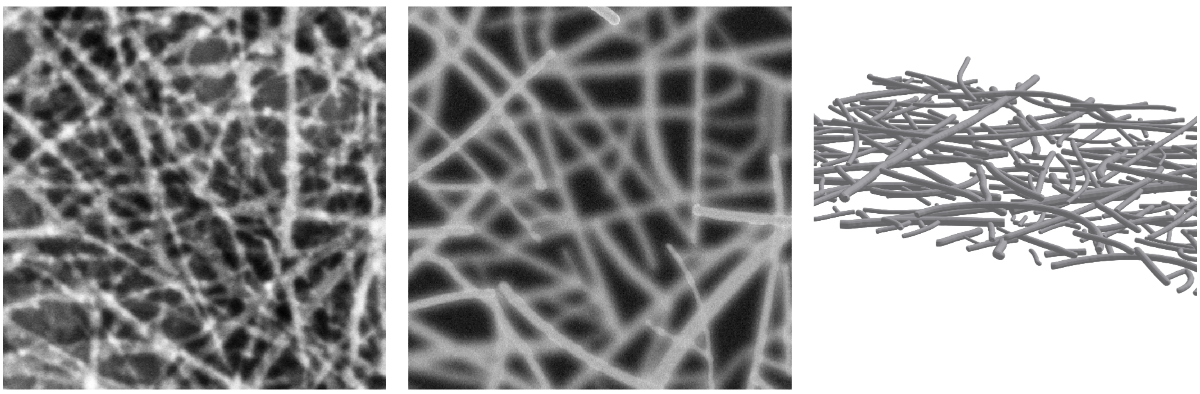

Figure 9. The same comparison on a real CNT SEM field, reconstructed from the axis of annotator A's annotation; the crop is 512 px across, about 0.55 µm of field B58_100. The depth order is not evaluated quantitatively here: real SEM provides no depth ground truth, since the annotation records which pixels belong to which strand, not which lies on top. The panels demonstrate the stage running end to end on real imaging; the forward model's appearance parameters are fitted to this micrograph.

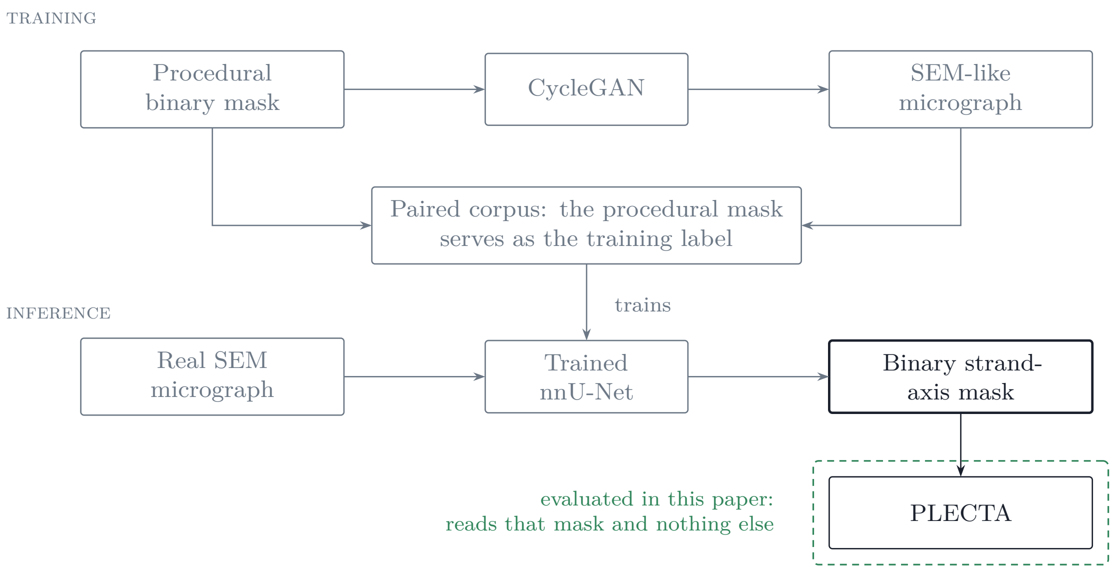

Figure 10. Upstream generation of the strand-axis mask. In training, image and mask pairs are generated procedurally and the procedural mask serves as the corresponding training label, so no annotator is involved in training; at inference the trained segmenter reads the micrograph, never an annotation, and ends in the binary strand-axis mask that is the whole interface to PLECTA. The segmenter is not a contribution of this paper and is not evaluated here; it carries its own evaluation [14], and only the stage in the dashed boundary is measured.

Table 3. Reconstruction from the manual-derived and the nnU-Net axis in three CNT SEM fields, under the measures of Section 3.6. *Coverage* is the fraction of the frame the full-width reference strands cover (Section 3.2), not the thin axis mask's. Detection F_1 is at IoU 0.1, tolerance 2 px.

Field	Coverage	Input axis	F_1	Join F_1	Det. F_1
B58_100	44.0 %	Manual-derived	0.89	0.93	0.94
		U-Net	0.46	0.56	0.63
B58_110	46.7 %	Manual-derived	0.84	0.91	0.93
		U-Net	0.45	0.59	0.60
B58_300	41.5 %	Manual-derived	0.95	0.96	0.96
		U-Net	0.47	0.62	0.58

axis, with annotator A's annotation as the reference in both cases, so the two conditions differ only in the mask. The first is an oracle-like input condition rather than an end-to-end result, asking what the grouping can do when the axis evidence is curated; only the second says what an automatic mask allows. Figure 11 shows both on one field, whole frame, with each input axis beside the corresponding PLECTA reconstruction.

Table 3 reports the three fields, per field and per condition. Averaged over them, PLECTA achieved F_1=0.89 from the manual-derived axis and 0.46 from the nnU-Net axis. The first condition evaluates grouping from curated real-image geometry, and establishes that the grouping holds on axis masks derived from real micrographs and not only on the generator's. The second quantifies sensitivity to an automatically generated mask, and is interpreted in Section 5 against what a second human annotator achieves on these same fields, which is less. 3 fields cannot estimate a population, and the inferential unit is the field.

The difference between the two conditions is attributable primarily to mask quality, and the errors consist predominantly of missed links rather than erroneous strand mergers. Beside annotator A's axis, the nnU-Net axis is a second, independent reading of the same strands, and it is thicker, broken into more pieces and displaced by a pixel or two almost everywhere, which panels (b) and (d) of Figure 11 show side by side. The consequence is fragmentation, which

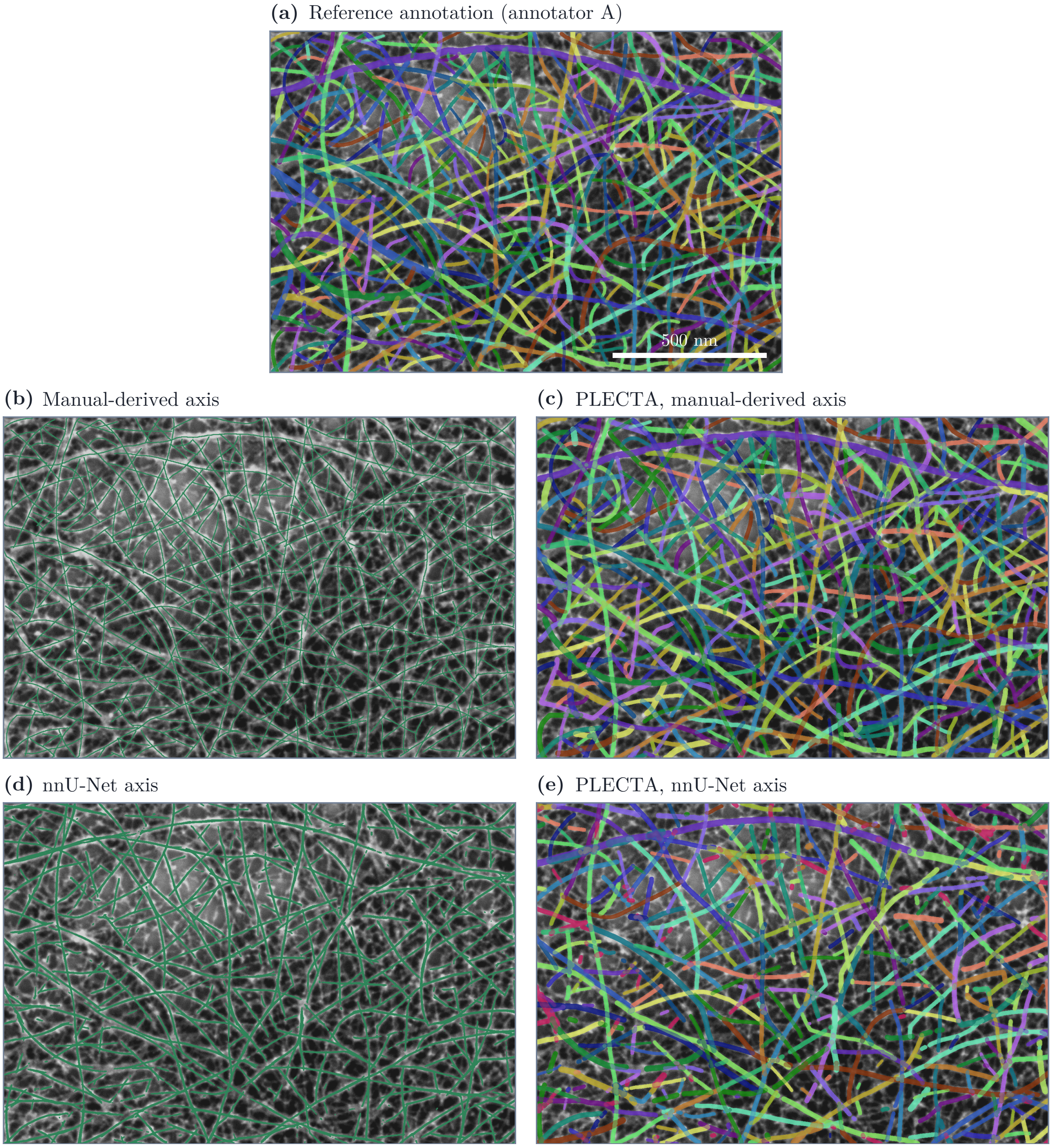

Figure 11. One CNT SEM field, B58_110, whole frame, a row per mask source; the scale bar in (a) is 500 nm, and every panel shows the same field. **(a)** the reference annotation (annotator A); **(b)** its skeleton, the manual-derived axis, and **(c)** PLECTA from it; **(d)** the segmenter's axis, and **(e)** PLECTA from that, same parameters. (b) and (d) are the input masks, in one colour, dilated a pixel for legibility; the rest are at width, through the layer of Section 3.5.1, which runs after the grouping is fixed and moves no pixel between strands. A predicted strand takes the colour of the annotated strand it matched; crimson matched none.

pairwise F_1 weights quadratically: a strand broken into several pieces is penalised once for every pair of pieces left unjoined. Join F_1, defined in Supplementary Section S3, assigns one error to each recoverable break instead; under it the difference between the conditions is reduced but not eliminated. The residual errors are predominantly missed links: on B58_110's nnU-Net axis, a join precision of 0.68 against a recall of 0.53, which panel (e) of Figure 11 shows.

These data do not permit separate estimation of segmentation error and inter-annotator ambiguity. Given annotator B's skeleton in place of the network's, PLECTA scores 0.38 against

Table 4. Component ablations and the two controls, on development scenes. The three blocks use different scene sets, so n is given per row and each ΔF_1 is against PLECTA on that row's own scenes; the column is not comparable across rows of different n.

Variant	n	F_1	ΔF_1	ARI
Fixed configuration	84	0.87	+0.000	0.86
No joint junction solve	30	0.78	-0.080	0.77
No gap bridging	84	0.73	-0.139	0.72
Single round	84	0.85	-0.020	0.84
No chain-extended frames	84	0.85	-0.021	0.84
No round schedule	84	0.86	-0.004	0.86
No curvature term	84	0.86	-0.005	0.86
No short-connector node merging	84	0.85	-0.022	0.84
No junction-cluster consolidation	84	0.80	-0.063	0.80
No junction chord-turn term	84	0.68	-0.186	0.67
Whole linker replaced	50	0.68	-0.145	0.67

annotator A, below what the network's axis reaches on every field and inside the 0.36–0.53 at which the two annotators agree without algorithmic processing. A second human reading of these strands is therefore no better an axis than the segmenter's, and the nnU-Net figure is bounded by how far one annotator's definition of a strand can be reproduced at all, which Section 5 examines.

4.4 Development ablations

Component contributions were evaluated by removing one component at a time and rescoring, together with two controls that are not components of the method. All ablations are measured on the development scenes.

Contribution of each component. Table 4 varies one thing at a time. Two rows change what a link costs, dropping the junction chord-turn term or the curvature term; two change the graph the linker reads, leaving neighbouring junction clusters unconsolidated or arms too short to merge the crossings they join; two change the iteration, running a single round or removing the schedule that relaxes the gates; and one removes gap bridging, so identity never crosses a break in the mask. The three largest losses are the junction chord-turn term, gap bridging and junction-cluster consolidation.

Two rows are controls rather than components. The greedy control retains all other PLECTA components but pairs each node's arms in ascending order of edge cost, rather than by whichever complete pairing of that node is cheapest overall; an early low-cost selection can preclude a later mutually compatible pairing, thereby causing fragmentation rather than erroneous merging. It also runs at parameters that were selected using the exact matcher, a bias in the exact matcher's favour. The last row replaces the linker outright, skeletonising the same mask and greedily pairing the branch ends that turn least inside fixed distance and angle gates, which is the pairing principle shared by SIFNE, GraFT's path cover and Wegmayr et al. [10, 35, 36].

4.5 Published methods on identical masks

PLECTA was compared with four published methods on two data sets: the 50 synthetic scenes the comparison was designed on, and the three annotated CNT SEM fields of Section 4.3, where the manual-derived axis lets the same comparison run on real imaging. All four are scored beside PLECTA on identical masks, through the same conversion and metric code, over scenes generated after every configuration was fixed. What is compared is narrow: the overlap-aware grouping of a binary axis mask of dense, crossing strands, at the coverages and tangling of the scenes described in Section 3.2. Each of these methods was built for a different setting, and how it performs in that setting is not what these numbers measure. DNAi [9], GraFT [10] and SIFNE [35] are run from their own released sources. For the fourth, Basu et al. [40], no implementation was found, and it is reimplemented here. Adapting each method to a common input and evaluation pipeline required methodological choices, and each of those choices is stated here.

Table 5. PLECTA against four published methods, identical masks and one scorer, ordered by F_1. The upper block is the degraded synthetic scenes; the lower is the annotated CNT SEM fields of Section 4.3 on the manual-derived axis, a plain mean over the three. Variation of information in bits, lower better; higher better for the rest.

* our reimplementation; no implementation of that paper was found. Supplementary Section S4 details it.

Method	F_1	Precision	Recall	ARI	VI total ↓	Crossing fidelity	Median s/scene at 20 %	Median s/scene at 60 %
Synthetic scenes, degraded axis								
PLECTA	0.82	0.83	0.81	0.82	0.57	0.81	0.14	4.09
SIFNE	0.71	0.77	0.67	0.70	0.93	0.77	3.42	5.68
Basu*	0.54	0.66	0.47	0.53	1.37	0.76	2.56	54.20
GraFT	0.49	0.67	0.40	0.48	1.56	0.68	2.44	29.74
DNAi	0.34	0.46	0.28	0.33	2.11	0.58	0.04	0.56
Real CNT SEM fields, manual-derived axis								
PLECTA	0.89	0.86	0.93	0.89	0.33	0.93		
SIFNE	0.87	0.86	0.88	0.86	0.37	0.92		
Basu*	0.76	0.74	0.78	0.76	0.59	0.89		

How each was run. Each released method has a front end that precedes the stage assigning identity, and in each case the mask is injected where that stage begins, so grouping is not confounded with segmentation and all identity-assignment stages run the original authors' code. Each was also run at a configuration selected on the same 84 development scenes PLECTA was configured on rather than at its own defaults. These adaptations favour the comparators, and by a wide margin: run end to end on the greyscale, GraFT scores 0.10 against 0.49 injected, and reconfiguration increased F_1 from 0.19 to 0.34 for DNAi and from 0.55 to 0.71 for SIFNE. Both are reported. Supplementary Section S4 gives each injection point, each selection, what was not attempted, and how each method fails.

Under these comparator-favouring settings, the ranking of Table 5 remained constant across density levels and mask conditions (Figure 12). SIFNE is the strongest of them, so the margin is measured against a method configured for this data rather than against an unsuitable default. Two entries qualify that ordering. The first is DNAi, whose deficit is a limitation of domain and density rather than a general ranking: it is built for DNA-fibre micrographs, which are sparse, and where the network is sparse and the axis clean it recovers 0.66 at 20% coverage. The second is Basu et al. [40], the closest structural relative PLECTA's junction matching has in that it too optimises over a neighbourhood and penalises joins by curvature. It stands second only to PLECTA on Crossing Fidelity while its pairwise F_1 is substantially below both released methods, so the curvature-priced matching is effective at the junctions, and the deficit arises in the stages around that decision. That is why the unmatched-arm penalty of Section 3.4 is claimed as part of a combination rather than on its own. Supplementary Section S4 gives both in full.

Each method in its own regime. That ordering is drawn from one generator, so it was tested in the regimes the two released methods were built for. On GraFT's own generator, run unmodified at its published settings over 40 scenes in 8 density strata, PLECTA achieved the highest F_1 at every density and no rank reversals occurred, including inside the range GraFT itself published, and the field-standard measures order the three identically. Overall it achieved F_1=0.92 there, against 0.48 for GraFT on its own scenes and 0.35 for DNAi. SIFNE released

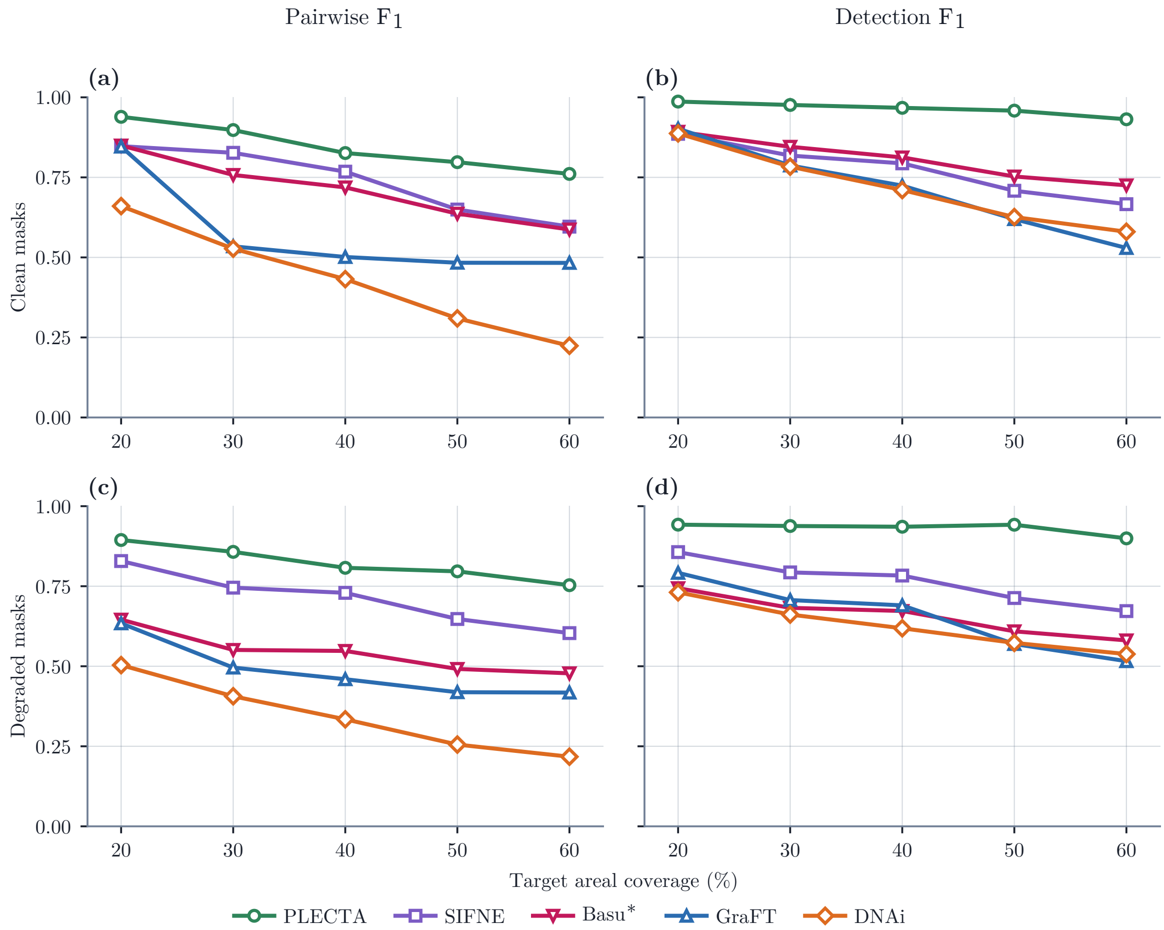

Figure 12. Five methods on identical masks, by coverage stratum. Rows: the undegraded axis, panels **(a)** and **(b)**, above the degraded one, panels **(c)** and **(d)**. Columns: pairwise F_1, which counts pairs of fragments, and detection F_1 at IoU 0.1 with a 2 px tolerance, which counts objects (Section 3.6). The two disagree where a method fragments, since only the first is quadratic in fragmentation. Crossing Fidelity is in Table 5. Means of ten scenes. * our reimplementation; none was found.

no generator, so the sparse cobweb topology its paper validates against was rebuilt here from that description; on that reconstruction the two methods are level in the regime the SIFNE publication addresses, 0.97 against 0.97, and separate only slightly at the denser end. That places the difference in the degree of crossing entanglement rather than in the total strand content of the frame, though a topology we rebuilt is weaker evidence than a generator its authors released. Supplementary Section S4 gives both density series in full.

Cost. The methods differ in computational cost as well as in accuracy. Per-scene cost was measured warm, single-threaded and pinned to one core, timing each method's own computation only (Table 5). These are independently optimised codebases, so absolute times measure implementation as much as algorithm; the transferable quantity is scaling, and the log–log exponents against foreground pixels are 2.11, 2.65 and 2.01.

On real imaging. The two comparators that read a binary axis were run over the three annotated fields as well, on the manual-derived masks of Section 4.3 and through the same scoring pipeline as PLECTA on those fields. Averaged over the three, PLECTA achieved F_1=0.89, against 0.87 for SIFNE and 0.76 for Basu, and the highest value on every measure reported. The ordering established on synthetic scenes therefore holds on masks read off real micrographs.

The margin is narrower than on the synthetic scenes, which likely reflects the quality of the manually derived axes: a skeleton of a hand annotation is unbroken, thin and placed where a person judged the strand to be, so it contains few of the fragmentation and displacement artefacts responsible for the differences observed in the synthetic evaluation.

5 Conclusions

PLECTA reconstructs overlapping strand instances from a binary strand-axis mask alone. It solves every junction, of any degree, as a single exact maximum-weight partial matching with an explicit penalty for unmatched arms, so a strand may terminate rather than receive a forced continuation, and a crossing pixel is assigned to every strand that passes through it.

- Over 128 held-out synthetic scenes from unseen seeds it achieved mean pairwise F_1=0.85 [0.84, 0.86], falling from 0.91 to 0.77 as areal coverage rises from 20 to 60 %.
- On identical masks it achieved the highest scores against four published methods at every density, and the ranking was preserved on a competitor's own generator, run unmodified at its published settings.
- The ablation analysis identified the junction chord-turn term and gap bridging as the largest contributors to reconstruction performance.
- A secondary stage reads the greyscale to order strands in depth. It is correct on 0.88–0.96 of the crossings it commits to, against a chance level of 0.5, and abstains where the image does not separate the pair by more than the local noise.
- The grouping core uses no random number generator and no training data, and its configuration is published, so identical input masks produce identical outputs.
- The end-to-end workflow does not require manually annotated training images: the upstream segmenter is trained on synthetic image and mask pairs in which the generating mask provides the label. The workflow therefore requires training, but not manual training annotations.

Scope and limitations.

- Within the generator, performance was insensitive to the evaluated strand-shape parameters: sweeping its bending stiffness over 50–150 % of default, and its curve wavelength over 10–40 px, moves the level means of F_1 by at most 0.05 and, respectively, 0.08, against a scene-to-scene scatter of 0.05 within a level (Supplementary Section S6).
- The comparison runs four methods on data none of them was built for. It supports a methodological claim: that pairing endpoints greedily, or committing with no option to decline, loses identity in dense crossings.
- Performance remains sensitive to errors in upstream segmentation. PLECTA can bridge a short omission, but it cannot infer a strand the mask omits, nor reliably reject a false axis introduced upstream.
- The real-material demonstration is a case study of three CNT SEM fields and cannot estimate generalisation. Scored against the annotator whose skeleton it was given, PLECTA achieved 0.79–0.95, and across annotators 0.37–0.51, while annotators A and B agree at 0.36–0.53 without algorithmic processing. The disagreement is systematic rather than error: B exports 1.85–2.83 times as many objects as A, shorter and far more fragmented, which reflects systematic differences in the operational definition of a strand instance. A real-data score is therefore bounded by that disagreement rather than by the method, and manual annotation cannot uniquely determine strand identity through locally ambiguous occlusions.
- Applicability is limited to networks in which directional continuity persists across the junction scale; reconstruction accuracy may decrease substantially when that assumption is violated, as for freely flexible chains.

Stronger conclusions would require an independent real-image test set defined before parameter selection, validated physical-width measurements, further released implementations evaluated under matched conditions, and an external-generator benchmark using previously unseen seeds.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at `https://github.com/Jorallan/PLECTA`. A graphical application that performs the same reconstruction on a supplied mask is available at `https://github.com/Jorallan/PLECTA_APP`. The annotation tool of Section 3.3 is at `https://github.com/Jorallan/ImageAnnotator`; it is listed because the machine-assisted annotation protocol described there is implemented in it. The manual annotations of the three SEM fields, from both annotators and in the form the scorer reads, will be deposited in 4TU.ResearchData. Generated scenes and detailed per-scene records are available from the corresponding author on reasonable request.

Acknowledgements

The authors thank the second annotator (annotator B of Section 3.3), who prefers to remain unnamed, for the independent re-annotation of 3 fields on which the reader-agreement bound in Section 5 rests. The software used is NumPy, SciPy, scikit-image, NetworkX and Matplotlib, and the authors thank their maintainers.

References

[1] Yuezhou Wang, Igor Ostanin, Cristian Gaidău, and Traian Dumitrică. Twisting carbon nanotube ropes with the mesoscopic distinct element method: Geometry, packing, and nanomechanics. *Langmuir*, 31(45): 12323–12327, 2015. doi: 10.1021/acs.langmuir.5b03208.

[2] Michael F. L. De Volder, Sameh H. Tawfick, Ray H. Baughman, and A. John Hart. Carbon nanotubes: Present and future commercial applications. *Science*, 339(6119):535–539, 2013. doi: 10.1126/science.1222453.

[3] Jonathan N. Coleman, Umar Khan, Werner J. Blau, and Yurii K. Gun'ko. Small but strong: A review of the mechanical properties of carbon nanotube-polymer composites. *Carbon*, 44(9):1624–1652, 2006. doi: 10.1016/j.carbon.2006.02.038.

[4] Alexey N. Volkov and Leonid V. Zhigilei. Structural stability of carbon nanotube films: The role of bending buckling. *ACS Nano*, 4(10):6187–6195, 2010. doi: 10.1021/nn1015902.

[5] Igor Ostanin, Roberto Ballarini, David Potyondy, and Traian Dumitrică. A distinct element method for large scale simulations of carbon nanotube assemblies. *Journal of the Mechanics and Physics of Solids*, 61(3): 762–782, 2013. doi: 10.1016/j.jmps.2012.10.016.

[6] Igor Ostanin, Traian Dumitrică, Sebastian Eibl, and Ulrich Rüde. Size-independent mechanical response of ultrathin carbon nanotube films in mesoscopic distinct element method simulations. *Journal of Applied Mechanics*, 86(12):121006, 2019. doi: 10.1115/1.4044413.

[7] Bernard K. Wittmaack, Abu Horaira Banna, Alexey N. Volkov, and Leonid V. Zhigilei. Mesoscopic modeling of structural self-organization of carbon nanotubes into vertically aligned networks of nanotube bundles. *Carbon*, 130:69–86, 2018. doi: 10.1016/j.carbon.2017.12.078.

[8] Oday Allan, Tanmaya Mishra, Robert Jan Meijer, and Matthijn de Rooij. Micro-CT analysis of geometrical distribution of filaments in double-twisted three-strand ropes: Comparison with analytical model. *Journal of Industrial Textiles*, 55:15280837251333116, 2025. doi: 10.1177/15280837251333116.

[9] Clément Playout, Yosra Mehrjoo, Renaud Duval, Marie Carole Boucher, Santiago Costantino, and Hugo Wurtele. DNAi: An open-source AI tool for unbiased DNA fiber analysis. *Nucleic Acids Research*, 54(7): gkag335, 2026. doi: 10.1093/nar/gkag335.

[10] Isabella Østerlund, Arne Neumann, Zhiming Ma, Yansong Miao, Staffan Persson, and Zoran Nikoloski. GraFT: A robust network-based spatiotemporal analysis of filamentous structures. *Science Advances*, 12(13): eadz4132, 2026. doi: 10.1126/sciadv.adz4132.

[11] Liyu Dong, Haibin Hang, Jin Gyu Park, Washington Mio, and Richard Liang. Detecting carbon nanotube orientation with topological analysis of scanning electron micrographs. *Nanomaterials*, 12(8):1251, 2022. doi: 10.3390/nano12081251.

[12] Tamjid Imtiaz, Jacques Doumani, Fuyang Tay, Natsumi Komatsu, Stephen Marcon, Motonori Nakamura, Saunab Ghosh, Andrey Baydin, Junichiro Kono, and Ahmed Zubair. Facile alignment estimation in carbon nanotube films using image processing. *Signal Processing*, 202:108751, 2023. doi: 10.1016/j.sigpro.2022.108751.

[13] Sahar Beigzadeh and Jeffrey E. Shield. Utilizing local orientation image analysis for microstructure quantification in additive manufacturing. *Materials Characterization*, 210:113761, 2024. doi: 10.1016/j.matchar.2024.113761.

[14] Bastian Rühle, Julian Frederic Krumrey, and Vasile-Dan Hodoroaba. Workflow towards automated segmentation of agglomerated, non-spherical particles from electron microscopy images using artificial neural networks. *Scientific Reports*, 11(1):4942, 2021. doi: 10.1038/s41598-021-84287-6.

[15] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28.

[16] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18:203–211, 2021. doi: 10.1038/s41592-020-01008-z.

[17] Tiberiu Stan, Zachary T. Thompson, and Peter W. Voorhees. Optimizing convolutional neural networks to perform semantic segmentation on large materials imaging datasets: X-ray tomography and serial sectioning. *Materials Characterization*, 160:110119, 2020. doi: 10.1016/j.matchar.2020.110119.

[18] Tsimur Davydzenka, Daniel Sinclair, Nikhilesh Chawla, and Pejman Tahmasebi. Deep-layers-assisted machine learning for accurate image segmentation of complex materials. *Materials Characterization*, 192:112175, 2022. doi: 10.1016/j.matchar.2022.112175.

[19] María Celeste Ramírez Trujillo, Teresa E. Alarcón, Oscar S. Dalmau, and Adalberto Zamudio Ojeda. Segmentation of carbon nanotube images through an artificial neural network. *Soft Computing*, 21(3):611–625, 2017. doi: 10.1007/s00500-016-2426-1.

[20] Nguyen P. Nguyen, Ramakrishna Surya, Matthew Maschmann, Prasad Calyam, Kannappan Palaniappan, and Filiz Bunyak. Self-supervised orientation-guided deep network for segmentation of carbon nanotubes in SEM imagery. In *Computer Vision – ECCV 2022 Workshops*, Lecture Notes in Computer Science, pages 412–428, 2023. doi: 10.1007/978-3-031-25085-9_24.

[21] Ivânia Maria Lucinda De Donato, Juliana do Nascimento Lunz, Fernanda Davi Marques, Bráulio Soares Archanjo, Francisco José Pereira Lopes, and Gilson Antônio Giraldi. Deep learning-based segmentation of nanomaterial images acquired via scanning electron microscopy. *Revista de Informática Teórica e Aplicada*, 32(1):83–90, 2025. doi: 10.22456/2175-2745.143516.

[22] Shuangmei He, Jiahao Tu, Yanlin Zhu, Weifu Li, Liming Tan, and Zi Wang. Instance segmentation and microstructure characterization based on powder microscopic image data. *Materials Characterization*, 234: 116051, 2026. doi: 10.1016/j.matchar.2026.116051.

[23] Carsen Stringer, Tim Wang, Michalis Michaelos, and Marius Pachitariu. Cellpose: A generalist algorithm for cellular segmentation. *Nature Methods*, 18:100–106, 2021. doi: 10.1038/s41592-020-01018-x.

[24] Uwe Schmidt, Martin Weigert, Coleman Broaddus, and Gene Myers. Cell detection with star-convex polygons. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Lecture Notes in Computer Science, pages 265–273, 2018. doi: 10.1007/978-3-030-00934-2_30.

[25] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371.

[26] Carsten Steger. An unbiased detector of curvilinear structures. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(2):113–125, 1998. doi: 10.1109/34.659930.

[27] Alejandro F. Frangi, Wiro J. Niessen, Koen L. Vincken, and Max A. Viergever. Multiscale vessel enhancement filtering. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI'98*, volume 1496 of *Lecture Notes in Computer Science*, pages 130–137, 1998. doi: 10.1007/BFb0056195.

[28] Yoshinobu Sato, Shin Nakajima, Nobuyuki Shiraga, Hideki Atsumi, Shigeyuki Yoshida, Thomas Koller, Guido Gerig, and Ron Kikinis. Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. *Medical Image Analysis*, 2(2):143–168, 1998. doi: 10.1016/S1361-8415(98)80009-1.

[29] Andrew M. Stein, David A. Vader, Louise M. Jawerth, David A. Weitz, and Leonard M. Sander. An algorithm for extracting the network geometry of three-dimensional collagen gels. *Journal of Microscopy*, 232 (3):463–475, 2008. doi: 10.1111/j.1365-2818.2008.02141.x.

[30] Jeremy S. Bredfeldt, Yuming Liu, Carolyn A. Pehlke, Matthew W. Conklin, Joseph M. Szulczewski, David R. Inman, Patricia J. Keely, Robert D. Nowak, Thomas R. Mackie, and Kevin W. Eliceiri. Computational segmentation of collagen fibers from second-harmonic generation images of breast cancer. *Journal of Biomedical Optics*, 19(1):016007, 2014. doi: 10.1117/1.JBO.19.1.016007.

[31] Ting Xu, Dimitrios Vavylonis, Feng-Ching Tsai, Gijsje H. Koenderink, Wei Nie, Eddy Yusuf, I-Ju Lee, Jian-Qiu Wu, and Xiaolei Huang. SOAX: A software for quantification of 3D biopolymer networks. *Scientific Reports*, 5:9081, 2015. doi: 10.1038/srep09081.

[32] Daniel A. D. Flormann, Moritz Schu, Emmanuel Terriac, Divyendu Thalla, Lucina Kainka, Marcus Koch, Annica K. B. Gad, and Franziska Lautenschläger. A novel universal algorithm for filament network tracing and cytoskeleton analysis. *The FASEB Journal*, 35(5):e21582, 2021. doi: 10.1096/fj.202100048R.

[33] Nathan A. Hotaling, Kapil Bharti, Haydn Kriel, and Carl G. Simon. DiameterJ: A validated open source nanofiber diameter measurement tool. *Biomaterials*, 61:327–338, 2015. doi: 10.1016/j.biomaterials.2015.05.015.

[34] Dorian Depriester, Sabine Rolland du Roscoat, Laurent Orgéas, Christian Geindreau, Benjamin Levrard, and Florian Brémond. Individual fibre separation in 3D fibrous materials imaged by X-ray tomography. *Journal of Microscopy*, 286(3):220–239, 2022. doi: 10.1111/jmi.13096.

[35] Zhen Zhang, Yukako Nishimura, and Pakorn Kanchanawong. Extracting microtubule networks from superresolution single-molecule localization microscopy data. *Molecular Biology of the Cell*, 28(2):333–345, 2017. doi: 10.1091/mbc.E16-06-0421.

[36] Viktor Wegmayr, Aytunc Sahin, Björn Sæmundsson, and Joachim M. Buhmann. Instance segmentation for the quantification of microplastic fiber images. In *2020 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 2199–2206, 2020. doi: 10.1109/WACV45572.2020.9093352.

[37] Yi Liu, Abhishek Kolagunda, Wayne Treible, Alex Nedo, Jeffrey Caplan, and Chandra Kambhamettu. Intersection to overpass: Instance segmentation on filamentous structures with an orientation-aware neural network and terminus pairing algorithm. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 125–133, 2019. doi: 10.1109/CVPRW.2019.00021.

[38] Max Frei and Frank Einar Kruis. FibeR-CNN: Expanding Mask R-CNN to improve image-based fiber analysis. *Powder Technology*, 377:974–991, 2021. doi: 10.1016/j.powtec.2020.08.034. Published online 2020.

[39] Yi Liu, Su Peng, Jeffrey Caplan, and Chandra Kambhamettu. Pick and trace: Instance segmentation for filamentous objects with a recurrent neural network. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, Lecture Notes in Computer Science, pages 635–645, 2023. doi: 10.1007/978-3-031-43993-3_61.

[40] Saurav Basu, Chi Liu, and Gustavo Kunde Rohde. Extraction of individual filaments from 2D confocal microscopy images of flat cells. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 12(3):632–643, 2015. doi: 10.1109/TCBB.2014.2372783.

[41] David Breuer and Zoran Nikoloski. DeFiNe: An optimisation-based method for robust disentangling of filamentous networks. *Scientific Reports*, 5:18267, 2015. doi: 10.1038/srep18267.

[42] Jaydeep De, Li Cheng, Xiaowei Zhang, Feng Lin, Huiqi Li, Kok Haur Ong, Weimiao Yu, Yuanhong Yu, and Sohail Ahmed. A graph-theoretical approach for tracing filamentary structures in neuronal and retinal images. *IEEE Transactions on Medical Imaging*, 35(1):257–272, 2016. doi: 10.1109/TMI.2015.2465962.

[43] Torben Peters, John Schumann, Kerstin Kämpf, and Asmus Meyer-Plath. AI-assisted recognition of fibre instances in scanning electron microscope images for aerosol characterization and exposure assessment. *Powder Technology*, 469:121909, 2026. doi: 10.1016/j.powtec.2025.121909.

[44] Cian Gabbett, Luke Doolan, Kevin Synnatschke, Laura Gambini, Emmet Coleman, Adam G. Kelly, Shixin Liu, Eoin Caffrey, Jose Munuera, Catriona Murphy, Stefano Sanvito, Lewys Jones, and Jonathan N. Coleman. Quantitative analysis of printed nanostructured networks using high-resolution 3D FIB-SEM nanotomography. *Nature Communications*, 15:278, 2024. doi: 10.1038/s41467-023-44450-1.

[45] Guillem Palou and Philippe Salembier. Monocular depth ordering using T-junctions and convexity occlusion cues. *IEEE Transactions on Image Processing*, 22(5):1926–1939, 2013. doi: 10.1109/TIP.2013.2240002.

[46] Hyunmin Lee and Jaesik Park. Instance-wise occlusion and depth orders in natural scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21178–21189, 2022. doi: 10.1109/CVPR52688.2022.02053.

[47] Menglei Chai, Lvdi Wang, Yanlin Weng, Yizhou Yu, Baining Guo, and Kun Zhou. Single-view hair modeling for portrait manipulation. *ACM Transactions on Graphics*, 31(4):116:1–116:8, 2012. doi: 10.1145/2185520.2185612.

[48] T. Y. Zhang and C. Y. Suen. A fast parallel algorithm for thinning digital patterns. *Communications of the ACM*, 27(3):236–239, 1984. doi: 10.1145/357994.358023.

[49] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.

[50] Marina Meilă. Comparing clusterings—an information based distance. *Journal of Multivariate Analysis*, 98(5):873–895, 2007. doi: 10.1016/j.jmva.2006.11.013.